

Module 5: Membranes - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Why do phospholipids arrange themselves into a bilayer in water?

- A. Their heads avoid contact with water.
- B. They are completely hydrophobic.
- C. Water-loving heads face outward while water-fearing tails tuck inward.
- D. They dissolve completely in water.
- Answer: C
- Why: The hydrophilic heads face the water and the hydrophobic tails hide inside, forming a stable two-layer membrane.

2. A small nonpolar molecule such as oxygen gas crosses the membrane mainly by:

- A. simple diffusion down its concentration gradient
- B. active transport that spends ATP
- C. not crossing at all
- D. osmosis of water
- Answer: A
- Why: Small nonpolar molecules pass directly through the bilayer from high to low concentration, with no energy required.

3. An animal cell is placed in pure water, a hypotonic solution. What happens?

- A. Water leaves and the cell shrinks.
- B. Nothing moves in or out.
- C. Salt rushes in and the cell shrinks.
- D. Water enters and the cell may swell or burst.
- Answer: D
- Why: In a hypotonic solution water moves into the cell by osmosis, so the cell swells and can burst.

4. Moving ions against their concentration gradient requires:

- A. only ordinary diffusion
- B. active transport that uses energy
- C. no membrane proteins
- D. removal of the membrane
- Answer: B
- Why: Movement against a gradient needs energy (ATP) and transport proteins, which is active transport.